Natural Language Processing
DSL504/DS504
Mid-Semester Examination
Time 2 hours

1. Find the edit distance between "house" and "homely" using the dynamic programming approach. Show the matrix of distances among prefixes. (5 Marks)

2. Consider the following simple PCFG:

S → NP VP	0.6	PropNoun → MUMBAI	0.2	
S → VP	0.4	PropNoun → RAJAT	0.2	
NP → NP PP	0.4	PropNoun → AYESHA	0.3	
NP → PropNoun	0.6	PropNoun → KOLKATA	0.3	
VP → Verb	0.3	Verb → TALK	0.5	
VP → Verb NP	0.3	Verb → MEET	0.5	
VP → VP PP	0.4	Prep → IN	0.4	
PP → Prep NP	1.0	Prep → WITH	0.6	

(a) Convert the above PCFG into Chomsky Normal Form.
(b) Use CKY parser to find the probability for generating the sentence: MEET RAJAT IN MUMBAI
(2+5 Marks)

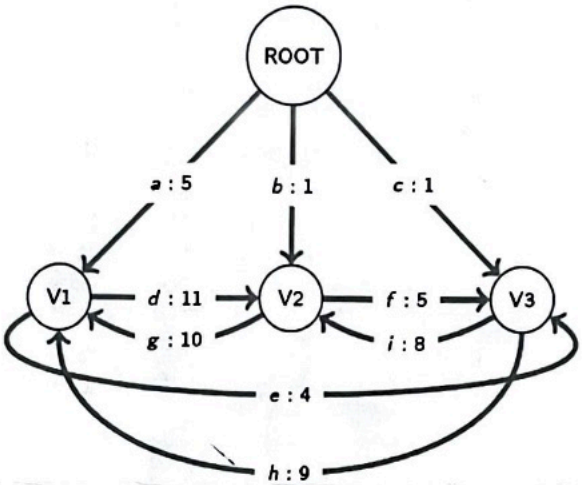
3. Suppose that you are given the following sentences
- Chinese Beijing Chinese
- Chinese Chinese Shanghai
- Chinese Macao
- Tokyo Japan Chinese

(a) Learn a Bi-gram language model using this data with add-1 smoothing.
(b) Using the language model learnt above, estimate the probability for the phrase, "Chinese Chinese Chinese Tokyo Japan" and calculate the corresponding perplexity score of the model for this phrase.
(4+4 marks)

4. Suppose you want to use a bigram POS tagger to tag the phrase, "the light watch", where we have been provided with the following probabilities:
P(Verb) = 0.2, P(Noun) = 0.3, P(Det) = 0.05, P(Adj) = 0.1,
P(the|Det) = 0.3, P(light|Noun) = 0.003, P(light|Adj) = 0.002, P(light|Verb) = 0.06, P(watch|Noun) = 0.003, P(watch|Verb) = 0.01,
P(Verb|Det) = 0.00001, P(Noun|Det) = 0.5, P(Adj|Det) = 0.3, P(Noun|Noun) =0.2, P(Noun|Adj) = 0.2, P(Noun|Verb) = 0.3, P(Verb|Noun) = 0.3, P(Verb|Adj) = 0.001, P(Verb|Verb) = 0.1

Work out in detail the steps of the Viterbi algorithm for performing POS tagging. Assume all other conditional probabilities which are not mentioned above, to be zero. (6 marks)

5. Consider the following graph with a root node and 3 other vertices. The edge weights between all the pairs of nodes have been provided. Use Chu-Liu-Edmonds algorithm to find the MST for this graph. You must clearly show all the steps. (6 marks)



6. a. Which problem does negative sampling solve for the skip-gram word2vec model? Explain using the corresponding objective function.

b. Suppose, you would like to predict the next word of the following phrase - "**The school is**" using the following two language models - i) a statistical/probabilistic language model and, ii) a RNN based language model. Given that this particular phrase is not present in the training data, will both the models work here? Justify.

c. In the RNN based language model, will "The" and "school" be given the same importance while predicting the fourth word of the given phrase? Explain using the equations of the model.

d. Mention advantages and disadvantages of RNN based language model over statistical language model. (3+2+2+3 marks)

7. a. Show the steps of transition based dependency parsing for the phrase "**He saw a comet with a telescope**" given the following dependency links (head => dependent): saw => He; saw => comet; saw => telescope; comet => a; telescope => with; telescope => a.

b. Now, assume the dependency tree is part of the training data for training a transition based dependency parser which uses the following three features - i) Is top of the stack a pronoun? (1/0) ii) Is top of the buffer a verb? (1/0) iii) Number of words between the top of the stack and top of the buffer. Show the data points which could be created from this particular training parse tree given above.

(5+3 Marks)